



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO PONNI FUELS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

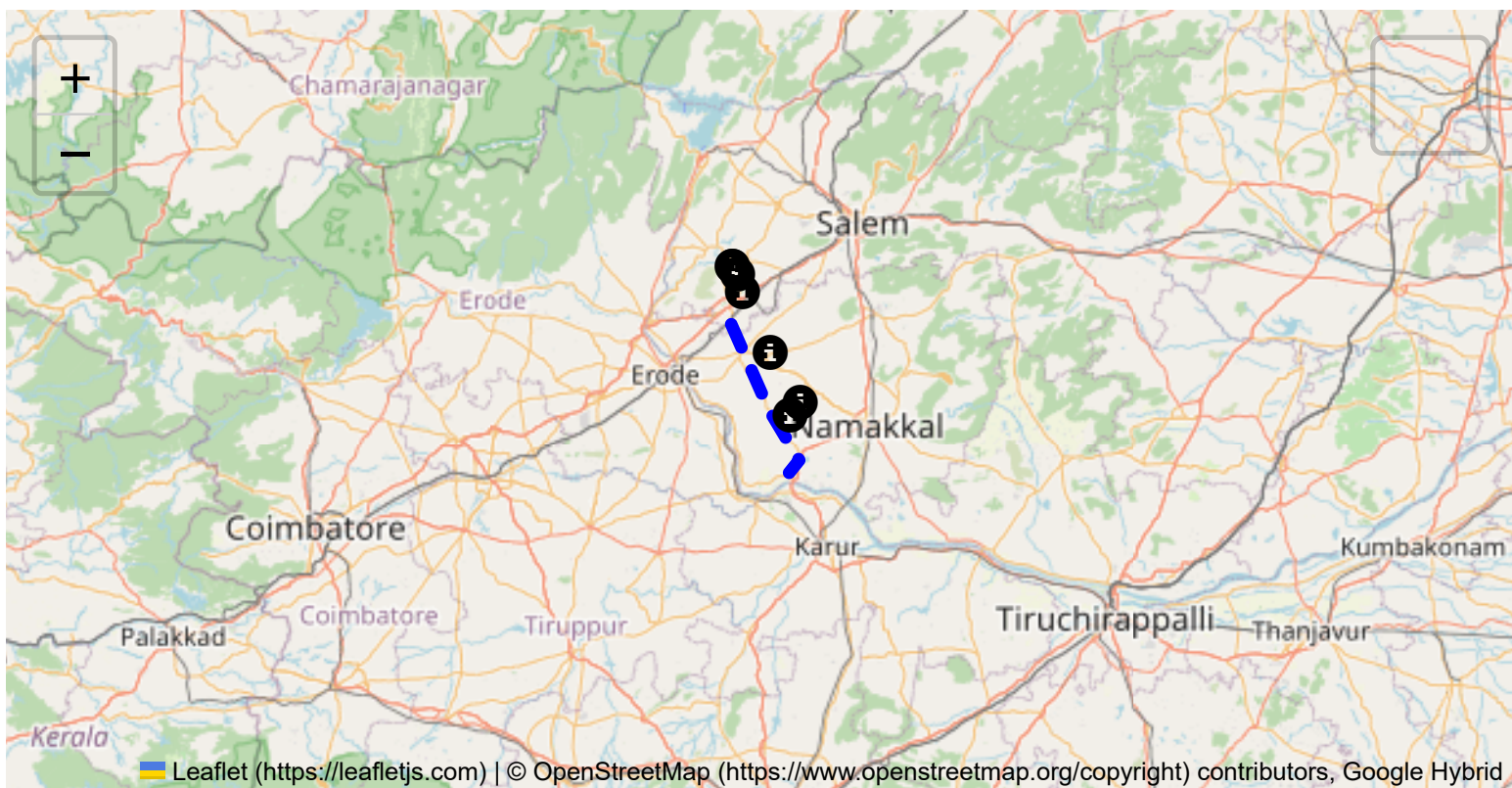
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

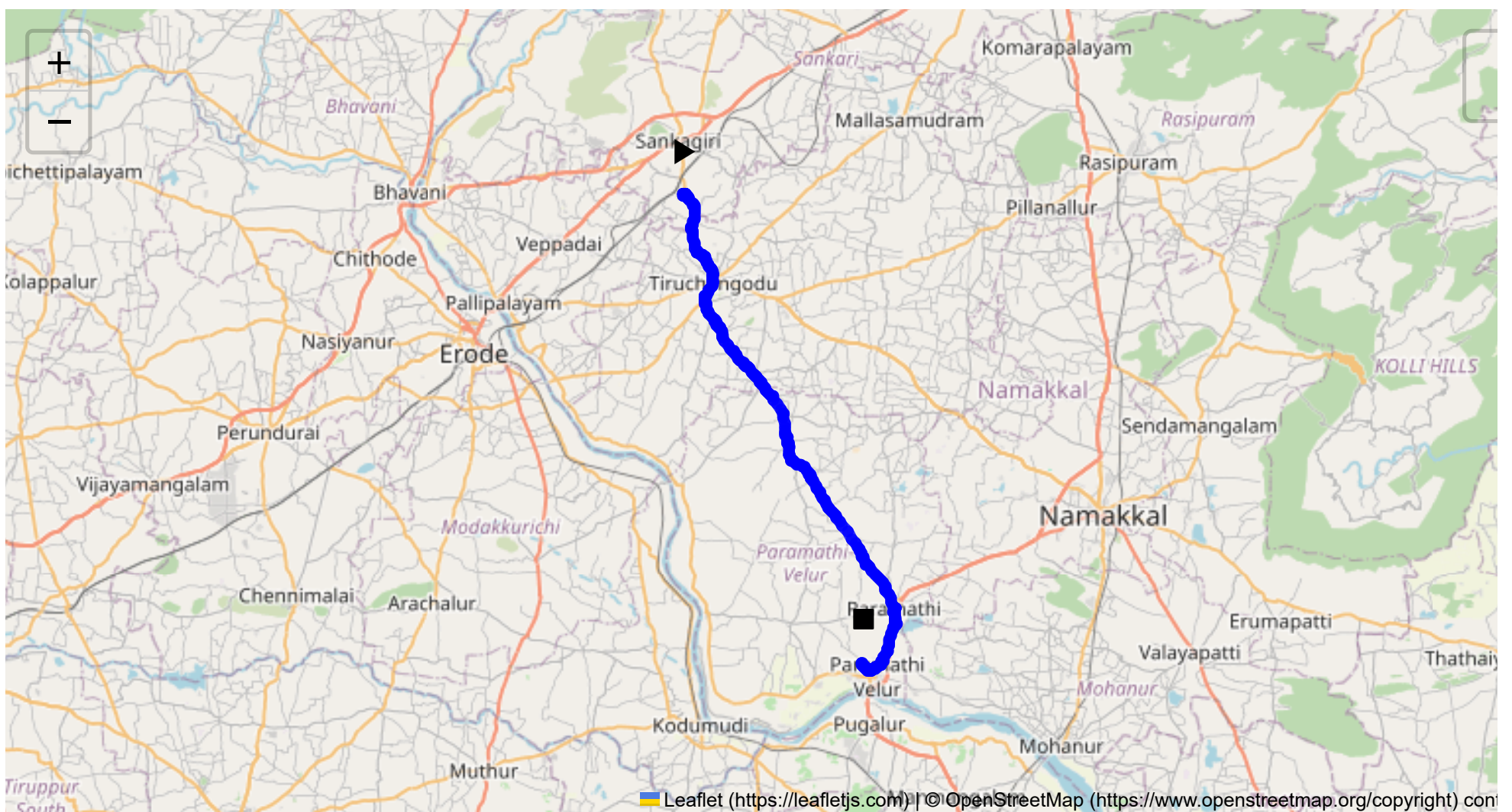
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 44.52 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.11785, 77.99815)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from CVQF+23W, Sangagiri to 4X9X+67, Tanipandal spans approximately 44.52 kilometers and involves traveling through several key locations: Puthur, Palmadai, Thondikaradu, Nallur, and Paramatti. This route predominantly follows local roads and State Highway 86, showcasing a mix of rural and semi-urban environments.

2. Typical Weather Conditions and Potential Hazards

Tamil Nadu generally experiences a tropical climate. The region usually faces high temperatures, especially between March and June. June to September is the monsoon season, which might result in heavy rains causing potential flooding or waterlogging on roads. Fog is occasional but can be an issue in the early mornings during winter months (December to February) that may affect visibility.

3. Traffic Patterns and Peak Hours

Traffic is moderate during regular hours with significant congestion near educational institutions, markets, and during early mornings and late evenings. Areas around Tiruchengode are particularly busier, especially near the Girls' High School where school timings can create congestion. Festivals and local events can additionally peak local traffic.

4. Road Quality and Infrastructure

The road quality varies along the route. State Highway 86 is relatively well-maintained but might have sudden patches of wear and tear. Most rural roads in this region are narrow, and unexpected potholes could present challenges. Regular maintenance checks are advisable for trucks to prevent accidents due to road conditions.

5. Alternative Routes for Emergencies

For emergencies, alternative routes may involve using NH544 and NH79, which provide better infrastructure and services. However, these are longer detours and might result in extended travel times.

6. Local Regulations for Hazardous Material Transport

Hazardous material transport in India is subject to road transport safety rules as outlined by the Petroleum and Explosives Safety Organisation (PESO). It's essential to follow all guidelines, including appropriate labeling and documentation. Restrictions may apply near sensitive areas like schools and hospitals.

7. Overview of Historical Incidents

While exact historical data is variable, incidents involving heavy vehicles on these roads often relate to poor weather conditions or road infrastructure failures. Sporadic reports of minor accidents due to the mix of heavy vehicles and local traffic occur.

8. Environmental Considerations and Sensitive Areas

Sensitive areas along the route include educational institutions and residential zones which require lower speeds and heightened caution. Cultural and religious sites can also mean increased pedestrian traffic.

9. Communication Coverage

Mobile network coverage in Tamil Nadu is generally reliable with few dead zones. However, there could be poor reception areas in rural segments around Palmadai and Nallur.

10. Estimated Emergency Response Times

Response times can vary, but typically, help from local town centers can arrive within 30 to 45 minutes. Response might be slower in more rural areas, suggesting a need for good emergency preparedness on behalf of drivers.

11. (Note: Originally numbered 12) Overall Summary of Risk Assessment

Overall, this route presents moderate risk primarily due to variable road conditions and potential weather impacts. Drivers should maintain caution during monsoon periods and be prepared for limited infrastructure support outside urban centers. Ensuring vehicles are well-maintained and communication devices are functional will reduce risks significantly. Additionally, adhering to local regulations and staying updated on weather forecasts will enable smooth transport of hazardous materials.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43961, 77.87341	30 KM/Hr
3	Turn	Medium	11.43968, 77.87345	30 KM/Hr
4	Blind Spot	Blind Spot	11.44029, 77.87544	10 KM/Hr
5	Turn	Medium	11.39671, 77.88878	30 KM/Hr
6	Turn	High	11.37868, 77.89498	15 KM/Hr
7	Turn	High	11.37850, 77.89453	15 KM/Hr
8	Turn	Medium	11.16764, 78.01565	30 KM/Hr
9	Turn	High	11.15573, 78.02176	15 KM/Hr
10	Turn	High	11.14545, 78.02208	15 KM/Hr
11	Blind Spot	Blind Spot	11.14480, 78.02291	10 KM/Hr
12	Turn	High	11.12555, 78.01664	15 KM/Hr
13	Turn	High	11.12540, 78.01613	15 KM/Hr
14	Turn	High	11.11211, 78.00450	15 KM/Hr
15	Turn	High	11.11780, 77.99813	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	T.C.A Hospital Tiruchengode	11.3791885, 77.8965774	30 km/h	Medium
6	clinic	Kongu Nursing Home	11.3783065, 77.8961134	30 km/h	Medium
7	hospital	Soorya Multispecialty Hospital	11.3786429, 77.8931912	30 km/h	Medium
8	hospital	Tiruchengode Government Hospital	11.37645, 77.89426	30 km/h	Medium
9	hospital	Krishna Hospital, Namakkal	11.3754811, 77.8931817	30 km/h	Medium
10	hospital	Tirukumaran Hospitals	11.3782118, 77.8914933	30 km/h	Medium
13	police	கரட்டுப்பாளையம் காவல் நிலையம்	11.357586, 77.8922539	30 km/h	Medium
14	hospital	Goverment Hospital, Nallur	11.2679698, 77.946863	30 km/h	Medium
16	hospital	Goverment Hospital	11.156647, 78.0213586	30 km/h	Medium
18	hospital	Indira Hospital, Namakkal	11.1119586, 78.0050232	30 km/h	Medium
20	hospital	Aachi Hospital	11.11017, 78.007066	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
11	school	KSR Educational institution	11.3772013, 77.8908807	30 km/h	Medium
12	school	MDV School	11.3719843, 77.8915244	30 km/h	Medium
15	school	Sivabakkiam Muthusamy Matriculation Higher Secondary School	11.1788177, 78.0061696	30 km/h	Medium
17	school	Goverment Higher Secondary School	11.1511724, 78.0200723	30 km/h	Medium
19	school	KKHSS	11.1108714, 78.0052766	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.39671, 77.88878



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37868, 77.89498



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.37850, 77.89453



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.16764, 78.01565



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.15573, 78.02176



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.14545, 78.02208



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.14480, 78.02291



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.12555, 78.01664



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.12540, 78.01613



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.11211, 78.00450
